# Pharmacology of the Adrenal Gland Part 2

Block 7 Independent Study

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Welcome to the 2nd part of the pharmacology of adrenal gland lecture. In the first part, we have covered deficiency of corticosteroids.

# Objective

Discuss the therapeutic options for the management of Cushing's syndrome with a focus on mechanism of action and side effects.

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In this part, we will focus on the pharmacological management of cushing's syndrome which is due to excess cortisol.

# Cushing's Syndrome

**Etiology**

- Pituitary disorders
  - Excessive pituitary ACTH production 2nd to microadenoma
  - ACTH dependent

- Adrenal disorders
  - Adenoma
  - ACTH independent

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- Etiology usually involves hyperactivity of HPA axis.
  - Pituitary disorders (60-70%)
    - Excessive pituitary ACTH production 2nd to microadenoma
      - ACTH dependent
  - Adrenal disorders (20-25%)
    - Autonomous secretion of cortisol by adrenal glands due to adenoma
      - ACTH independent

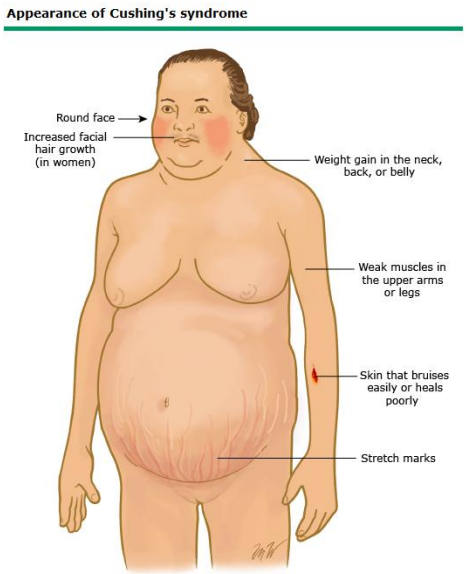
## Cushing's Syndrome

1. **Ectopic Cushing's Syndrome**
2. **Iatrogenic**

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- Two types of Cushing's syndrome.
- Ectopic Cushing's Syndrome
  - Tumors outside of the HPA axis produce an ACTH-like substance
  - Complete suppression of pituitary ACTH
  - ACTH dependent
- Iatrogenic
  - ACTH independent
  - long-term administration of glucocorticoid in supraphysiological doses

# Clinical Signs and Symptoms




https://www.uptodate.com/contents/cushings-syndrome-beyond-the-basics/print

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"moon face", "buffalo hump", central obesity, weakness/fatigue, purplish abdominal striae, bruising, osteopenia, depression, hypertension, glucose intolerance
Excess ACTH: Skin hyperpigmentation
Excess androgen secretion:Acne, hirsutism, thinning of the scalp hair, amenorrhea

# Nonpharmacological Therapy

- Surgery
- Radiation
- Bilateral adrenalectomy

**Patient will only live 4-5 years if untreated (CAD, DM, electrolyte)**

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Surgery

Cure rates of 80-90%
Macroadenomas' cure rate is < 50%
Recurrence rate is 10-30%

Radiation (2nd line therapy)

Cure rate is 80%
Used for mild disease
Patient failed surgery

Bilateral adrenalectomy

Required chronic glucocorticoid and mineralcorticoid replacement

## Pharmacological Therapy

- Adjunctive therapy
- Disease usually relapses when medications are discontinued
- Increased adverse effects with prolonged use

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Useful in weak patients, inoperable tumors, or awaiting/following radiation therapy

# 1. Steroidogenic Inhibitor

- **Mitotane**
  - MOA
    - Adrenolytic agent—exerts cytotoxic effects on adrenal cortex
    - Medical adrenalectomy (complete ablation of cortisol production)
- Adverse effects
  - GI
    - Nausea/anorexia/diarrhea
  - Neuromuscular
    - Fatigue/muscle weakness
    - CNS depression (somnolence/dizziness)

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## Mitotane is

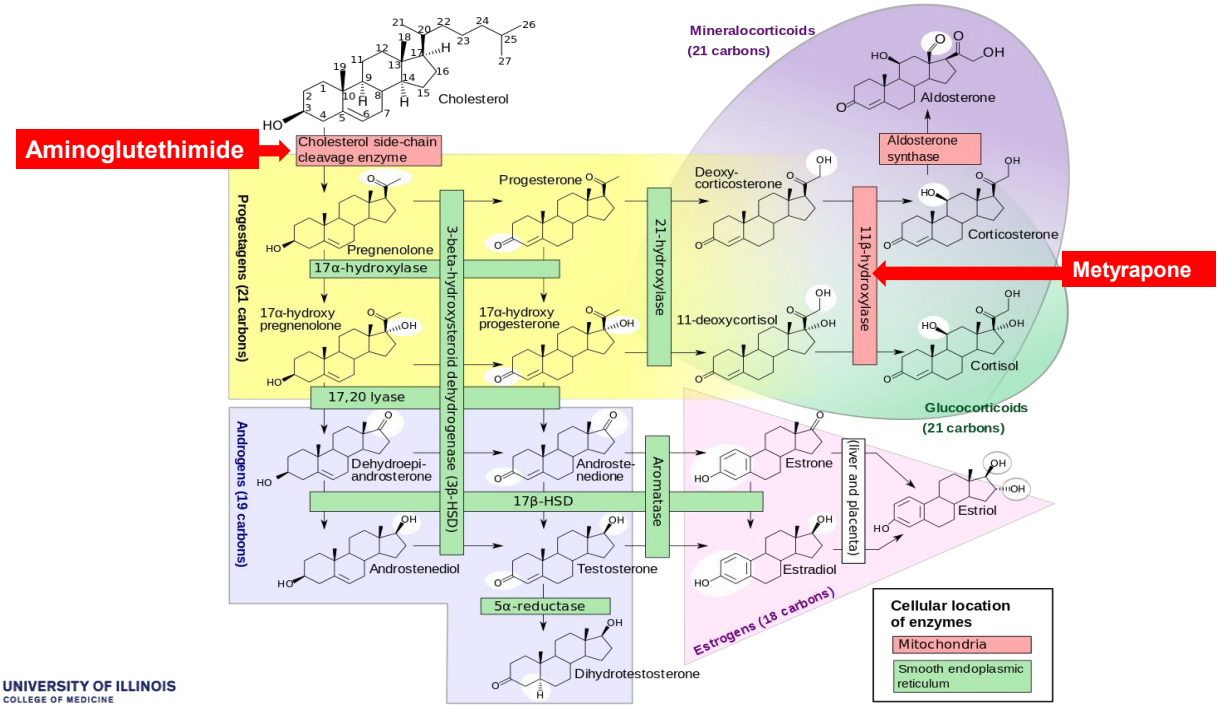
MOA

Adrenolytic agent—exerts cytotoxic effects on adrenal cortex
Medical adrenalectomy (complete ablation of cortisol production)
Long term maintenance glucocorticoid therapy
Slow in onset of action – over weeks

Initiate with 1-6 grams/day in divided doses
Duration of therapy is 6-9 months
Monitor urinary free cortisol and adjust dose as needed

# 2. Adrenal Enzyme Inhibitors

Aminoglutethimide
Metyrapone
Ketoconazole




https://commons.wikimedia.org/w/index.php?curid=4494399

# Aminoglutethimide

MOA

- Inhibits cholesterol side chain cleaving enzyme (cholesterol desmolase)

Adverse reactions

- Drowsiness, bone marrow suppression, drug-induced lupus

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Overall, reduces the synthesis of adrenal glucocorticoid, mineralocorticoid, estrogen, aldosterone, and androgens

# Metyrapone

MOA
- Inhibits steroid 11 Beta-hydroxylase
- Used in combination with other enzyme inhibitor

- Adverse effects
  - Hypertension, headache, lightheadedness, rash, hirsutism

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Inhibits steroid 11 Beta-hydroxylase
Final step in cortisol biosynthesis
May also inhibit ACTH secretion at high doses

Clinical data is limited (compassionate use).
Compassionate drug use (or sometimes just *compassionate use*) is the use of a new, unapproved drug to treat a seriously ill patient when no other treatments are available.

# Ketoconazole

MOA

- Inhibited steroid 17 alpha-hydroxylase, 11 beta-hydrozylase, and cholesterol demolase at higher doses,
- potent inhibitor of testosterone

Adverse effects

- ⇑ LFTs,
- Gynecomastia,
- Decreased libido,
- Impotence
- GI symptoms

- Drug interactions are plenty due to its enzyme inhibiting effects
- Monitor liver enzymes

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Ketoconazole is an antifungal agent,
Liver func test

# 3. Glucocorticoid Receptor Antagonist

Mifepristone

- MOA
  - Progesterone and glucocorticoid receptor antagonist
- Side effects
  - Fatigue, GI, headache, HTN, endometrial thickening, hypokalemia
  - Abortifacient agent



Abortifacient agent—should not be used in pregnant or women wishing to become pregnant

# Question

Blocking which one of the following is the most effective way to inhibit adrenal hormone synthesis from cholesterol?

A. 17-alpha hydroxylase

B. 11 Beta-hydroxylase

C. Cholesterol desmolase